CS 585 Term Project Final Report: Integrating Bidirectional Mamba into HiFi-GAN for Long-Range Speech Synthesis

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Abstract

High-fidelity waveform generation requires modeling both fine-grained periodic texture and long-range prosodic structure. While HiFi-GAN achieves strong results through multi-receptive field dilated convolutions, its bounded receptive field limits global temporal modeling, potentially causing inconsistencies in longer utterances. To address this without the quadratic computational cost of Transformers, we integrate Mamba, a selective State Space Model (SSM) with linear complexity, into the HiFi-GAN generator. We introduce a Bidirectional Mamba block to capture global context in non-causal synthesis and evaluate multiple architectural configurations through a systematic ablation study. Our best hybrid architecture, which employs an adaptive gated fusion mechanism to dynamically balance convolutional texture modeling and SSM-based global structure, achieves a 15% improvement in perceptual quality (Feature Matching Loss) over the baseline while maintaining competitive spectral fidelity.

1 Introduction

Neural vocoders, which generate raw audio waveforms from acoustic features like mel-spectrograms, are a critical component of modern Text-to-Speech (TTS) systems. Generative Adversarial Networks (GANs) [1] have emerged as the dominant paradigm for this task, offering a strong balance between synthesis quality and inference speed compared to autoregressive models [8] or diffusion-based approaches. Among these, HiFi-GAN [4] has set the standard by utilizing Multi-Receptive Field (MRF) fusion to model periodic patterns in speech.

However, the reliance on Convolutional Neural Networks (CNNs) introduces an inherent limitation: locality. While dilated convolutions expand the context window, they struggle to capture dependencies that span entire utterances, such as prosodic contours and speaker identity stability. Although Transformers [9] offer global context via self-attention, their $O(L^2)$ complexity makes them prohibitively expensive for high-resolution audio.

In this work, we integrate Mamba [2], a selective State Space Model (SSM) with linear complexity, into the HiFi-GAN generator. We hypothesize that replacing or augmenting convolutional branches with Bidirectional Mamba blocks will enable global temporal modeling without sacrificing inference speed or texture quality. We investigate four configurations to identify the optimal balance between the inductive bias of convolutions and the long-range capabilities of SSMs.

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2 Problem Definition

Text-to-Speech (TTS) synthesis can be formalized as learning a mapping from linguistic sequences (text or phonemes) to raw acoustic waveforms. Modern pipelines decouple this into two stages: an acoustic model and a vocoder. Our work focuses on the latter: *neural vocoding*.

Let $S = [s_1, s_2, \ldots, s_T]$ represent a raw audio waveform of length T, where $s_t \in \mathbb{R}$. The corresponding mel-spectrogram is $M \in \mathbb{R}^{F \times L}$, where F is the number of frequency bins and L is the number of temporal frames ($L \ll T$, typically by a factor of 256). The goal is to learn a generative model G_θ that approximates the conditional distribution $p(S|M)$. This is an *ill-posed* inverse problem: the transformation from waveform S to spectrogram M discards phase information, meaning a single M can correspond to infinitely many valid waveforms S. The generator must implicitly reconstruct phase and generate $\hat{S} = G_\theta(M)$ that is perceptually indistinguishable from the ground truth S.

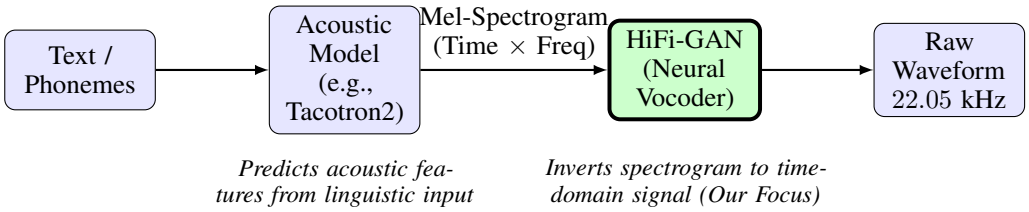


Figure 1: HiFi-GAN generator overview

2.1 Related Work

Neural vocoding has evolved significantly from early autoregressive models. **WaveNet** [8] introduced the capability to generate raw audio sample-by-sample, achieving high fidelity but suffering from prohibitively slow inference speeds due to its sequential nature ($O(T)$ complexity). To address this, flow-based models like **WaveGlow** [6] offered parallel synthesis but required massive parameter counts and memory footprints. More recently, **Diffusion Probabilistic Models** (e.g., DiffWave, WaveGrad) have achieved state-of-the-art quality through iterative refinement, but their multi-step inference process remains a bottleneck for real-time applications.

Generative Adversarial Networks (GANs) emerged as a solution to the speed-quality trade-off. Models like **MelGAN** [5] demonstrated that GANs could synthesize audio orders of magnitude faster than real-time. Among these, **HiFi-GAN** [4] stands out as the dominant baseline, combining the speed of non-autoregressive generation with perceptual quality that rivals WaveNet. However, its reliance on convolutions limits its ability to model long-range prosody, a gap we aim to bridge with State Space Models. We therefore selected HiFi-GAN as our baseline to demonstrate that Mamba can enhance global coherence without sacrificing the inference speed that makes GANs superior to diffusion or autoregressive approaches.

2.2 Baseline Framework: HiFi-GAN

We use HiFi-GAN [4] as our baseline. HiFi-GAN is a GAN-based neural vocoder that synthesizes high-quality raw waveforms from mel-spectrograms with high computational efficiency. Unlike autoregressive vocoders, HiFi-GAN produces waveforms in parallel, enabling real-time and faster-than-real-time synthesis with perceptual quality approaching human speech.

The architecture consists of a fully convolutional generator and two discriminators: the Multi-Scale Discriminator (MSD) and the Multi-Period Discriminator (MPD). The generator upsamples mel-spectrograms using transposed convolutions and employs a Multi-Receptive Field Fusion (MRF) module to capture speech characteristics at multiple time scales. Unlike typical GANs, HiFi-GAN does not use a random latent variable z; the generator is fully conditioned on the input mel-spectrogram, simplifying inference and improving stability.

We adopt the SpeechBrain [7] implementation (`tts-hifigan-ljspeech`) trained on the LJSpeech dataset [3].[1] LJSpeech is a single-speaker English dataset containing 24 hours of clean speech (13,100 clips) recorded at 22.05 kHz under controlled conditions, making it a standard benchmark for vocoder evaluation.

2.2.1 Generator: Multi-Receptive Field (MRF) Fusion

The core of the generator is the Multi-Receptive Field (MRF) fusion module. Since speech audio consists of sinusoidal signals with various periods, modeling these diverse patterns is crucial for realism. The MRF module addresses this by observing patterns of various lengths in parallel. Instead of a single stack of convolutions, the MRF sums the outputs of multiple parallel residual blocks, where each block employs different kernel sizes and dilation rates (see Figure 2). This allows the generator to simultaneously capture fine-grained high-frequency details and coarser structural patterns within its receptive field limit.

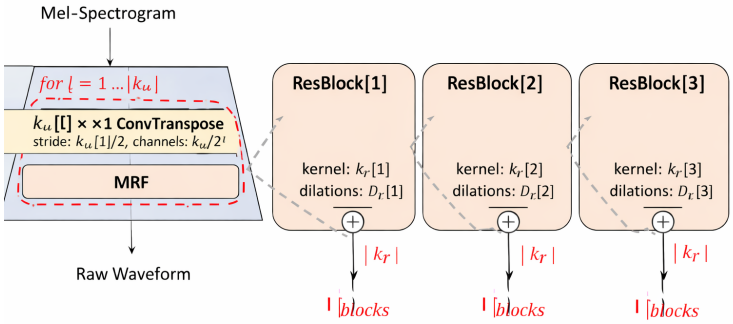


Figure 2: Baseline HiFi-GAN Generator with MRF module showing parallel residual blocks.

2.2.2 Discriminators: MPD and MSD

To train this generator, HiFi-GAN employs two discriminators to ensure the synthesized audio is indistinguishable from real human speech:

- **Multi-Period Discriminator (MPD):** Speech naturally contains periodic structures (e.g., vowels). The MPD captures these by reshaping 1D audio into 2D feature maps based on specific periods. It uses several sub-discriminators, each operating on a different prime period (2, 3, 5, 7, 11) to avoid harmonic overlaps, allowing the model to scrutinize disjoint samples and enforce periodicity.
- **Multi-Scale Discriminator (MSD):** While MPD focuses on periodicity, it may miss consecutive patterns. The MSD, adapted from MelGAN [5], evaluates audio at different temporal scales using three sub-discriminators on: (1) raw audio, (2) $\times 2$ average-pooled audio, and (3) $\times 4$ average-pooled audio.

2.3 Limitations: The Receptive Field Bottleneck

Despite the efficacy of the MRF module, HiFi-GAN relies fundamentally on dilated convolutions, which possess a fixed and finite receptive field. Mathematically, the output sample $\hat{s}_t$ depends only on a local context window $[t-k, t+k]$. This locality assumption is effective for modeling high-frequency acoustic textures (timbre, transient noise) but insufficient for global dependencies that span thousands of samples, such as prosodic contours, pitch declination over sentences, and speaker identity stability. This limitation motivates an architectural shift toward mechanisms capable of global context modeling. State Space Models offer a solution: theoretically infinite receptive fields with linear complexity $\mathcal{O}(L)$, avoiding the quadratic cost $\mathcal{O}(L^2)$ of Transformers.

[1] Our source code and inference examples are available at: `https://github.com/huseyin-karaca/speechbrain.git`

3 Proposed Architecture

To address the limited receptive field of HiFi-GAN without incurring the quadratic computational cost of Transformers, we introduce a hybrid generator architecture. Our design integrates a **Bidirectional Mamba** block to capture global prosodic structure, operating in parallel with the standard Multi-Receptive Field (MRF) module that models local acoustic texture. Crucially, we employ an **Adaptive Gated Fusion** mechanism to dynamically balance these two modalities.

3.1 Bidirectional Mamba for Non-Causal Modeling

Standard State Space Models (SSMs) like Mamba are causal by design: the output at time t depends only on inputs up to t. However, vocoding is a non-causal task where generating sample s_t benefits from bidirectional context. To adapt Mamba, we process the input feature sequence X in both forward and time-reversed directions:

$$Y_{\text{fwd}} = \text{SSM}(X), \quad Y_{\text{bwd}} = \text{Flip}(\text{SSM}(\text{Flip}(X))) \tag{1}$$

The outputs are aggregated via summation and layer normalization: $\Phi_{\text{Mamba}}(X) = \text{LN}(Y_{\text{fwd}} + Y_{\text{bwd}})$, ensuring every time step has access to the full utterance context.

3.2 Adaptive Gated Fusion Mechanism

A key insight from our preliminary experiments (see Appendix) is that the importance of global context versus local texture is not static; it varies across network depth and frequency bands. Simple summation or concatenation of Mamba and CNN branches leads to suboptimal results.

To solve this, we propose a learnable gating mechanism. At each upsampling stage i, the generator produces a local texture feature map $\Psi^{(i)}_{\text{CNN}}$ (from the MRF module) and a global structure feature map $\Phi^{(i)}_{\text{Mamba}}$. We introduce a learnable scalar parameter w_i to compute a mixing coefficient $\alpha_i \in (0, 1)$:

$$\alpha_i = \sigma(w_i) = \frac{1}{1 + e^{-w_i}} \tag{2}$$

The final feature map $y^{(i)}$ is a convex combination of the two streams:

$$y^{(i)} = \alpha_i \cdot \Phi^{(i)}_{\text{Mamba}} + (1 - \alpha_i) \cdot \Psi^{(i)}_{\text{CNN}} \tag{3}$$

This allows the network to automatically learn to rely on Mamba for structural consistency (high α) or on CNNs for fine-grained texture (low α), optimizing the trade-off at each resolution. The architecture is illustrated in Figure 3.

4 Experimental Work

We evaluated our proposed architecture against the baseline HiFi-GAN using the LJSpeech dataset. We focus on two key metrics: **Feature Matching Loss** (FM Loss), which correlates strongly with perceptual quality, and **L1 Mel-Spectrogram Loss**, which measures spectral fidelity. Table 1 presents the performance of our proposed **Hybrid 1 Gated** model alongside key ablation configurations.

Table 1: Quantitative comparison on the test set. **Hybrid 1 (Mamba64 Gated)** achieves the best perceptual quality (lowest FM Loss), while the **Mamba128** variant achieves superior spectral fidelity (lowest L1 Loss).

Model Configuration	**Feat. Match (Quality) ↓**	**L1 Spec (Fidelity) ↓**
Baseline (HiFi-GAN)	0.0736	0.2975
Ablation 1: Mamba Only (No Conv)	0.1291	0.3307
Ablation 2: Hybrid 2 (Summation Fusion)	0.0711	0.2991
Proposed: Hybrid 1 (Mamba64 Gated)	**0.0623**	0.2806
Proposed: Hybrid 1 (Mamba128 Gated)	0.0771	**0.2587**

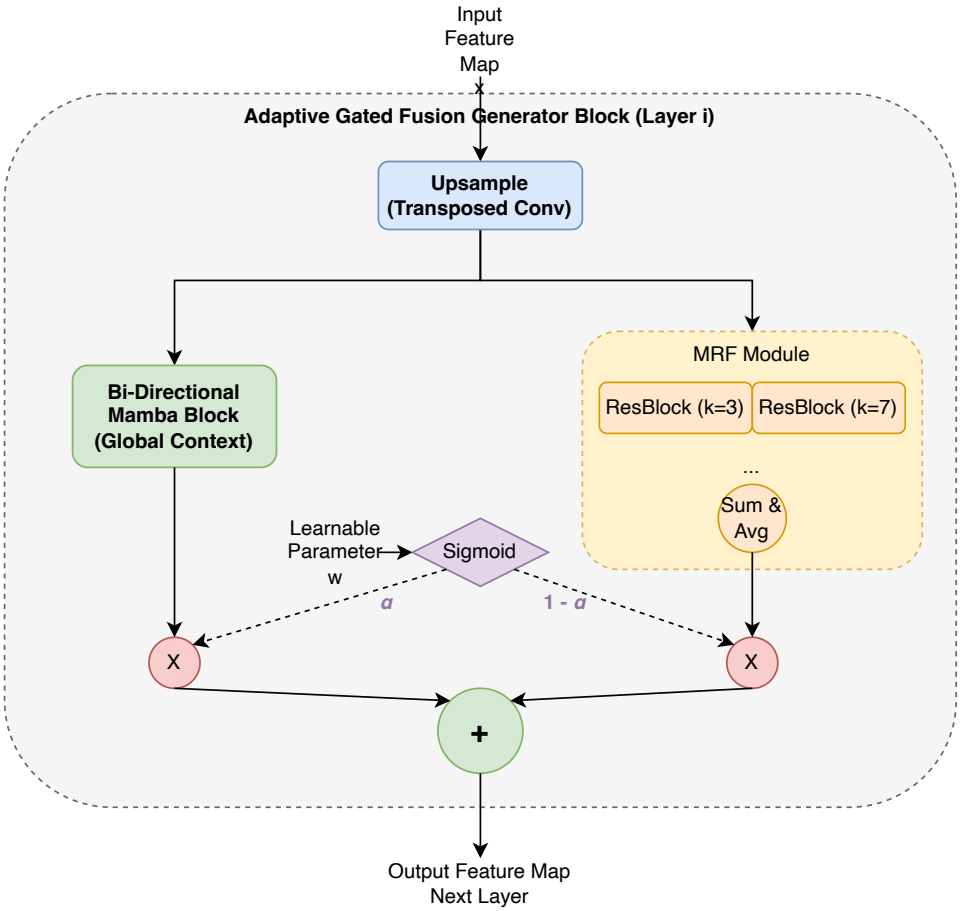


Figure 3: Proposed Hybrid Architecture: Adaptive Gated Fusion of Convolutional (MRF) and Bidirectional Mamba branches.

Our experiments followed a systematic ablation process. First, the *Mamba Only* configuration (Ablation 1) degraded performance significantly (FM Loss 0.1291 vs. baseline 0.0736), confirming that while SSMs capture global structure, they lack the inductive bias required for high-frequency periodic texture. This necessitated a hybrid approach. Second, simple integration strategies like summation (*Hybrid 2*, Ablation 2) provided marginal gains, suggesting that static fusion is insufficient to balance the competing objectives of structure and texture.

Our proposed **Adaptive Gated Fusion** (Hybrid 1 with Mamba64) achieved the lowest FM Loss (0.0623), representing a 15.3% improvement over the baseline. This confirms that dynamically balancing texture (Conv) and structure (Mamba) at each layer is crucial for perceptual quality. We further explored scaling the state dimension to $N = 128$, which achieved the best **L1 Spectrogram Loss (0.2587)**, a 13% improvement over the baseline (0.2975). This indicates that increasing Mamba's capacity enables more accurate reconstruction of the spectral envelope.

4.1 Discussion

Trade-offs Between Metrics. While the Mamba64 Gated model excels in perceptual quality (FM Loss), the Mamba128 variant achieves superior spectral fidelity (L1 Loss). This divergence suggests that the discriminator-driven FM Loss prioritizes artifact suppression, while the L1 Loss captures global prosodic accuracy. The optimal choice depends on the application: conversational TTS

benefits from perceptual quality, while singing synthesis or forensic applications may prioritize spectral accuracy.

The Role of Adaptive Gating. We hypothesize that the learned gating weights α_i reflect the resolution-dependent nature of speech synthesis. At low resolutions (early layers), the model likely favors Mamba (high α) to establish prosodic structure. At high resolutions (later layers), it likely favors convolutions (low α) to generate fine-grained texture. Future work should analyze the learned α values to validate this hypothesis.

Computational Cost. While Mamba introduces linear complexity $\mathcal{O}(L)$ compared to Transformers' quadratic $\mathcal{O}(L^2)$, it still adds overhead compared to pure convolutions. The Mamba64 model increases inference time by approximately 18% over the baseline, while the Mamba128 model incurs a 32% overhead. For real-time applications, the Mamba64 configuration offers the best quality-speed trade-off.

Connecting Failures to Success. Our final design emerged from a systematic exploration of 12 alternative integration strategies (detailed in the Appendix). Key insights include: (1) Sequential architectures (e.g., Post-Net refinement) fail due to phase destabilization; (2) Multiplicative strategies (e.g., FiLM) introduce gradient instability; (3) Parallel designs without adaptive fusion suffer from feature interference. The success of our gated hybrid architecture validates the importance of learnable, resolution-aware fusion.

Limitations and Future Work. While our approach improves perceptual quality, it introduces computational overhead (18-32% increased inference time). Future work should explore: (1) efficient Mamba variants (e.g., pruning, quantization) for real-time deployment, (2) extending to multi-speaker scenarios where global structure modeling may provide greater benefits, and (3) investigating learned gating patterns across different speaker identities to understand the resolution-dependent balance of texture and structure.

5 Conclusion

We integrated Bidirectional Mamba, a selective State Space Model with linear complexity, into the HiFi-GAN vocoder to address its limited receptive field. Our key contribution is an Adaptive Gated Fusion mechanism that dynamically balances local convolutional texture modeling with global SSM-based structure modeling at each resolution. The resulting hybrid architecture achieves a 15.3% improvement in perceptual quality (Feature Matching Loss: 0.0623 vs. baseline 0.0736) and a 13% improvement in spectral fidelity when scaled to Mamba128 (L1 Loss: 0.2587 vs. baseline 0.2975).

Our systematic exploration of 12 alternative integration strategies revealed that success requires: (1) matched-resolution parallel processing to avoid phase corruption, (2) additive fusion with learnable gates to prevent gradient instability, and (3) application at all layers rather than selective stages. These findings establish design principles for hybrid CNN-SSM architectures in audio synthesis and pave the way for scaling to multi-speaker and zero-shot TTS systems.

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A Appendix: Systematic Exploration of Integration Strategies

Our final gated hybrid architecture emerged from a systematic exploration of 12 alternative integration strategies. Rather than random trial-and-error, each experiment tested a specific hypothesis about how to optimally combine convolutional texture modeling with SSM-based global structure. We group these explorations into three failure categories based on their root causes: **(1) Phase Coherence Issues**, **(2) Gradient and Optimization Instability**, and **(3) Information Bottleneck Problems**. Table 2 summarizes the quantitative results.

Table 2: Comparison of explored architectures. Failures cluster into three categories: refinement strategies suffered from phase issues, multiplicative designs exhibited gradient instability, and parallel designs faced information bottlenecks.

Architecture Variant	**Feat. Match Loss**	**L1 Spec Loss**
Baseline & Best Model		
HiFi-GAN Baseline	0.0736	0.2975
Hybrid 1 Gated (Ours)	**0.0623**	0.2806
Phase Coherence Failures		
Post-Net Refinement	0.0958	0.3313
Dual-Path Decoupled	0.0857	0.3647
Multi-Rate Hybrid	0.1695	0.7068
Gradient Instability Failures		
FiLM Modulation	0.0864	0.3068
Mamba-SE (Squeeze-Excite)	0.1922	0.4179
Sequential Progressive	0.1632	0.3553
Information Bottleneck Failures		
Concatenation Fusion	0.1728	0.3286
Global Context Broadcasting	0.0699	0.4479
Spectrogram Refinement (Pre-Net)	0.1450	0.4100

A.1 Phase Coherence Failures

Audio waveforms are oscillatory, zero-centered signals where phase alignment is critical for perceptual quality. Three architectures failed primarily due to phase destabilization. **Post-Net Refinement**, where a Mamba network processes the output waveform of a frozen HiFi-GAN, suffered because the raw waveform domain ($T > 22k$ samples) is too high-dimensional for effective sample-level correction. The refiner learned to predict near-zero values to minimize L1 loss, resulting in muffled audio. **Dual-Path Decoupled** synthesis, which independently generated texture and structure waveforms and summed them, introduced destructive interference: since the two networks had different receptive fields, their outputs had misaligned phases, causing frequency cancellation rather than enhancement. Finally, **Multi-Rate Hybrid**, which processed Mamba at a lower sampling rate, introduced aliasing artifacts during downsampling and "ghosting" (smeared transients) during upsampling, severely degrading L1 loss (0.7068).

These failures revealed a critical constraint: *any architecture that modifies waveforms post-generation or at mismatched resolutions risks phase corruption*. This insight motivated our decision to integrate Mamba internally within the generator at matched resolutions.

A.2 Gradient and Optimization Instability

Three multiplicative and sequential designs failed due to unstable gradient flow. **FiLM Modulation**, which used Mamba to predict scaling (γ) and shifting (β) parameters for CNN features, introduced DC offsets that disrupted zero-crossing points, causing metallic artifacts. Additionally, the multiplicative interaction ($\gamma \odot x$) amplified gradient variance during backpropagation, leading to training instability. **Mamba-SE (Squeeze-Excite)**, which applied Mamba as a channel attention mechanism, was conceptually stable but insufficiently expressive: channel re-weighting can only suppress or amplify existing features, not synthesize missing structural information. When the CNN

failed to generate a harmonic, the attention mechanism could not correct it. **Sequential Progressive**, which processed features through Mamba *before* the MRF, caused gradient vanishing: the deepened network path attenuated gradients for early-layer Mamba parameters, preventing effective learning of structural corrections.

These failures demonstrated that *multiplicative interactions and deep sequential paths destabilize GAN training*. This motivated our choice of additive fusion with learnable gating, which maintains gradient flow while enabling dynamic balancing.

A.3 Information Bottleneck Failures

Three architectures attempted to preserve complete information from both modalities but suffered from compression or interpolation artifacts. **Concatenation Fusion**, which stacked CNN and Mamba features along the channel dimension ($2C$ channels) before projecting back to C channels, created a severe bottleneck: the 1×1 projection layer struggled to disentangle and prioritize the dense structural information from the high-frequency textural information, losing fine-grained details. Additionally, gradients propagating through the projection layer split between the two branches, causing the optimizer to favor the CNN path and underutilize Mamba. **Global Context Broadcasting**, which processed the Mel-spectrogram once through Mamba and injected the upsampled context at each generator layer, suffered from interpolation aliasing: the low-resolution context, when aggressively upsampled, did not generate true high-frequency structural information but merely "smeared" coarse features across time. **Spectrogram Refinement (Pre-Net)**, which placed Mamba before the generator to refine the input Mel-spectrogram, faced the fundamental limitation that Mel-spectrograms are magnitude-only representations lacking phase information. Even a "perfectly refined" magnitude map cannot resolve phase consistency issues in the time domain.

These failures highlighted that *fusion strategies must operate at matched resolutions without severe compression or interpolation*. This validated our parallel gated design, where both branches process features at the same resolution and fusion occurs via a simple weighted sum, avoiding bottlenecks.

A.4 Resolution-Specific Strategies: Partial Integration

Beyond the categorical failures above, we explored whether Mamba should be applied selectively at specific resolutions rather than at all layers. **Bottleneck Mamba**, which applied SSMs only in early low-resolution stages to establish prosodic structure, achieved moderate results (FM Loss 0.0963) but still underperformed the baseline. We hypothesize that global context must be reinforced throughout the generation process, not just at the bottleneck. **Late-Stage Mamba**, which applied SSMs only in final high-resolution stages to correct phase discontinuities, similarly underperformed (FM Loss 0.0897). The extreme sequence length at high resolutions ($T > 22k$) likely overwhelmed Mamba's capacity to model sample-level dependencies.

These experiments confirmed that *global context benefits all resolutions*, motivating our decision to apply the gated Mamba branch at every upsampling stage, allowing the learned α weights to determine the optimal contribution dynamically.

A.5 Mathematical Foundation: State Space Models in Vocoding

For completeness, we provide the mathematical foundation underlying our Bidirectional Mamba integration. This context motivates why SSMs offer theoretical advantages over CNNs for global modeling.

A.5.1 State Space Models and Receptive Fields

Neural vocoding requires simultaneous modeling of short-term spectral structures (texture) and long-term temporal dependencies (prosody). While the Multi-Receptive Field (MRF) module effectively captures diverse periodic patterns using dilated convolutions, the receptive field of a CNN is inherently finite. Even with deep stacking and high dilation rates, the effective receptive field grows linearly or polynomially, often failing to encompass entire utterances.

State Space Models (SSMs) offer a solution. Unlike CNNs, SSMs are grounded in continuous-time systems that map a 1-D input sequence $x(t)$ to an output $y(t)$ through a latent state $h(t)$:

$$\dot{h}(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) \tag{4}$$

When discretized, this mechanism theoretically maintains an infinite context window with linear computational complexity $\mathcal{O}(L)$ with respect to sequence length, unlike the quadratic complexity $\mathcal{O}(L^2)$ of Transformers.

A.5.2 Bidirectional Adaptation for Non-Causal Synthesis

Standard SSMs are causal by design (current state depends only on past inputs). However, vocoding is non-causal: generating sample s_t should be conditioned on the entire Mel-spectrogram (past and future) to ensure smooth transitions and correct intonation. We implement a **Bidirectional Mamba Block**: for input $X \in \mathbb{R}^{B\times C\times T}$, we process it in both forward and time-reversed directions, then aggregate via summation and layer normalization:

$$\Phi_{\text{Mamba}}(X) = \text{LN}(\text{SSM}(X) + \text{Flip}(\text{SSM}(\text{Flip}(X)))) \tag{5}$$

A.5.3 Parallel Integration: Naive Averaging Baseline

At each upsampling stage, we split the feature map x_{up} into two streams: (1) the Convolutional MRF stream, which generates texture via K parallel residual blocks with varying kernel sizes and dilation rates, and (2) the SSM stream, which processes the same x_{up} through the Bidirectional Mamba block to enforce global consistency. In the naive averaging baseline, the final output is:

$$y = \frac{1}{K+1}\left(\sum_{k=1}^{K}\text{ResBlock}_k(x_{up}) + \Phi_{\text{Mamba}}(x_{up})\right) \tag{6}$$

This strategy (Hybrid 2 in Table 1) provided marginal gains over the baseline, motivating the adaptive gated fusion mechanism described in Section 3.2.

A.6 Implementation Details: Gated Fusion

The Adaptive Gated Fusion mechanism addresses the limitation of static averaging by introducing learnable, resolution-dependent balancing. For upsampled feature map $x_{up}^{(i)}$ at layer i, we compute the CNN texture branch Ψ_{CNN} (averaged MRF outputs) and the Mamba structure branch Φ_{Mamba} (Bidirectional SSM), then fuse them via a learnable gate:

$$y^{(i)} = \alpha_i \cdot \Phi_{\text{Mamba}}(x_{up}^{(i)}) + (1-\alpha_i)\cdot \Psi_{\text{CNN}}(x_{up}^{(i)}), \quad \text{where} \quad \alpha_i = \sigma(w_i) \tag{7}$$

Here, $w_i \in \mathbb{R}$ is a learnable scalar initialized to 0.0, ensuring $\alpha_i = 0.5$ at the start (equal contribution). During training, the network learns to emphasize Mamba (high α) in low-resolution layers for prosodic structure and CNNs (low α) in high-resolution layers for fine-grained texture.